Center Estates ERP

CONSTRUCTION SUITE & MOBILE APP

Eliminating Capital Leakage in African Real Estate Development.

Real-time field telemetry, BoQ allowance shields, and cryptographic milestone stage-gates.

VERIFY ON-SITE → CERTIFY ON WEB

Live Demo Project: Karam Apartments — Lang'ata (PRJ-KRM-01)

EXECUTIVE SUMMARY

A Single Operating Chain Between Site Gate & Bank Drawdown

- Bill of Quantities (BoQ) Integrated: Every material delivery checked against contractual budget limits.
- Contractual Stage-Gates: Downstream milestones remain locked until prerequisite structural works reach 100%.
- Offline-First Site Diary: Complete Clerk of Works logging with zero 4G/Wi-Fi reliance via local SQLite sync.
- Tamper-Evident Auditing: SHA-256 evidence seals, GPS geofencing, and biometric digital sign-offs.
- Mini-BoQ Valuations: Real-time calculation of Interim Payment Certificates (IPCs) backed by verified site photos.

INDUSTRY CHALLENGE

The Three Critical Blind Spots in Construction Oversight

Why 15% to 25% of commercial construction capital is lost during execution.

⚠️ 1. Phantom & Over-Deliveries

Paper delivery slips signed at the gate with no live cross-reference to the approved BoQ allowance.

- Materials delivered beyond budget without developer visibility.
- Inability to track which supplier provided which batch.
- Discrepancies only discovered months later during final accounts.

⛔ 2. Premature Progress Claims

Contractors claiming 40% completion on superstructure before foundation works are certified.

- Downstream milestone payments disbursed prematurely.
- Capital over-extended ahead of actual physical progress.
- Subcontractor disputes and compromised structural verification.

📄 3. Disconnected Site Diary

Clerk of Works field diaries written in paper logbooks or scattered WhatsApp groups.

- Photos lack GPS coordinates and tamper-evident timestamps.
- Site defects and safety snags unaddressed for weeks.
- Disconnect between site reality and the finance office.

AUDIT PRINCIPLE **Traditional oversight depends on trust and retrospective audits. Center Estates enforces proactive digital controls.**

SYSTEM ARCHITECTURE

Unified Real-Time Platform: Mobile App ↔ Web ERP ↔ API

One continuous operational record linking the site gate directly to the board room.

Mobile Field App (Expo / React Native)

Authoritative source of physical site truth for Clerks of Works and resident engineers.

- Offline-First: TinyBase reactive layer on local SQLite.
- Rugged Field UX: Big touch targets, high contrast, Kenyan supplier chips.
- GPS Geofencing: Automatic classification of photo capture location.

Central REST API (Django & MySQL)

Production backend engine running at api.centerestates.com.

- Bidirectional Delta-Sync: Atomic /sync/pull and /sync/push endpoints.
- Contractual Stage-Gate Validator: Enforces 100% prerequisite completion.
- Cryptographic Integrity Seals: SHA-256 evidence hashing.
- Role-Based Access

Office Web ERP (Next.js 14)

Real-time management command center for Developers, PMs, and Quantity Surveyors.

- Work Plan & CPM Gantt: Predecessor scheduling and progress.
- Bill of Quantities (BoQ): Line-item budgets, rates, and variance shields.
- Mini-BoQ Valuations: Certified interim payment calculations.
- Defect Kanban: Live

AUDIT PRINCIPLE **Mobile evidence is immutable. Submitted site records cannot be overwritten or deleted by office users.**

FIELD MATERIAL CONTROL

Site Gate Deliveries Bound to BoQ Allowances

Preventing material leakage before the truck leaves the site perimeter.

🔍 Searchable BoQ Picker

Clerk taps to open full project Bill of Quantities. Auto-fills item code, material name, and measurement unit.

🏢 Kenyan Supplier Chips

One-tap selection for Bamburi, Apex Steel, Devki, Savannah, Simba, and Tononoka.

🛡️ Planned Allowance Shield

Displays approved BoQ quantity and remaining balance. Red banner triggers if delivery exceeds planned allowance.

🏷️ Work Plan Task Tagging

Ties materials to CPM work plan activities (e.g. WP-001 Foundation Works) for exact consumption tracking.

- Eliminates handwritten gate passes: Delivery note numbers and supplier invoices digitized at intake.
- Automatic reconciliation: Real-time variance reports alert Quantity Surveyors to unexpected material spikes.
- Offline caching: Form drafts automatically persist in SQLite even if phone battery dies.

AUDIT PRINCIPLE **Every delivery record is linked to a BoQ line item and a work plan activity, ensuring 100% material traceability.**

CONTRACTUAL GOVERNANCE

The Stage-Gate Engine: Mathematical Payout Protection

Downstream milestones cannot advance until preceding structural requirements reach 100%.

Substructure & Raft
(Prerequisite)

100% COMPLETE
✓ Certified by Engineer

→

Superstructure Frame
(Current Milestone)

STAGE-GATE UNLOCKED
Progress Claim Allowed

→

Roofing & Waterproofing
(Downstream)

🔒 LOCKED
0% Allowed until Frame 100%

- Enforced at Database & API Level: The backend rejects progress updates (>0%) if predecessors are incomplete.
- Prominent Mobile Banner: The Clerk's app explicitly states which predecessor milestones are blocking submission.
- Protects Developer & Financier: Eliminates contractor over-billing on downstream finishes before structural sign-off.

AUDIT PRINCIPLE **Stage-gates bridge engineering reality with legal contract compliance, removing subjective certification bias.**

QUALITY & COMPLIANCE

Defect Management & Concrete Lifecycle Schedulers

Logging site snags on mobile; scheduling 7, 14, and 28-day concrete crushing tests.

⚡ Mobile Snags & Defect Ticketing

Resident Engineers log defects instantly from the field.

- Classification: Quality Defect, Safety Risk, Structural Discrepancy.
- Priority Routing: Urgent, High, Normal.
- Photographic Evidence: Direct camera capture with GPS coordinates.
- Live Office Kanban: Real-time sync to the web

🧱 Concrete Quality & Curing Scheduler

Ensuring structural pour compliance with Kenyan building standards (BS 8110 / Eurocode 2).

- Slump Test Logging: Real-time workability verification.
- Automated Crushing Schedule: Automatically queues 7-day, 14-day, and 28-day cylinder crushing tests.
- Lab Certificate

milestone record.

🦺 Health & Safety Incident Reports

Immediate escalation protocol for near-misses and site incidents.

- Severity grading and root cause categorization.
- Corrective Action Plan (CAP) tracking.
- Regulatory reporting compliance (DOSHS & NCA).

AUDIT PRINCIPLE **Quality tests and defect tickets form part of the permanent structural audit trail required for building occupation certificates.**

COMMERCIAL OVERSIGHT

Interim Payment Certificates (IPCs) & Mini-BoQ Audits

Turning certified field progress into automated financial drawdown certificates.

1. Mini-BoQ Valuation

Quantity Surveyors inspect the exact BoQ sub-elements linked to the completed milestone, calculating certified KES value.

2. AI Progress Audit

AI models compare actual progress against baseline schedule, highlighting material variance and projected completion dates.

3. Drawdown Certificates

Generates audit-ready PDF payment certificates showing certified value, retention deductions, and advance recovery.

- 100% Backed by Field Evidence: Every shilling certified is backed by geotagged site photos and delivery notes.
- Automatic Retention Deductions: Configurable retention rates (e.g. 5% or 10%) automatically deducted.
- Disbursement Reconciliation: Seamless financial handover to property development accounts.

AUDIT PRINCIPLE **Quantity Surveyors save up to 80% of interim valuation time while providing banks with unassailable audit proof.**

ANTI-FRAUD ARCHITECTURE

Offline Resilience & Cryptographic Anti-Fraud Controls

Built specifically for African construction environments with unreliable connectivity.

📶 Offline SQLite Engine

Basements and deep excavations have zero 4G.

The app caches all forms, BoQ items, and daily logs locally in SQLite, syncing automatically upon network restoration.

👆 Biometric Key Signatures

Each field device enrols an Ed25519 cryptographic key in hardware keystore.

Submissions require a live biometric prompt, providing non-repudiation of who signed the milestone.

📍 GPS Geofence Guard

Photos captured off-site are flagged as fraud risks.

The server automatically tests hardware GPS coordinates against the project perimeter.

🔒 Immutable SHA-256 Seals

Every submission generates a tamper-evident hash over progress, observations, and photo bytes.

Any posterior alteration breaks the seal.

AUDIT PRINCIPLE **Security and auditability are baked into the data layer, not added as an afterthought.**

DEMONSTRATION FLOW

Recommended 5-Step Live Demonstration Script

A compelling 8-minute journey demonstrating real-time office-to-field synchronicity.

Step 1	**Office Baseline Setup (Web)**	Open Karam Apartments. Showcase the Work Plan (CPM Gantt) and approved BoQ allowances.	**1.0 min**
Step 2	**Site Gate Delivery (Mobile)**	Open Mobile App. Select BoQ item (Cement), tap 'Bamburi' chip, demonstrate Planned Allowance Shield.	**2.0 mins**
Step 3	**Stage-Gate Lockout (Mobile)**	Open 'Superstructure Frame'. Show '🔒 Stage-Gate Locked' banner blocking premature claim.	**1.5 mins**
Step 4	**Live Snag Sync (Mobile → Web)**	Log rebar spacing defect on mobile. Switch to Web to show immediate appearance on Kanban board.	**1.5 mins**
Step 5	**Mini-BoQ Valuation & AI Audit (Web)**	Open Mini-BoQ breakdown, verify certified KES amount, and trigger AI Progress Audit.	**2.0 mins**

AUDIT PRINCIPLE **Keep the audience focused on Karam Apartments to prove coherent end-to-end operational integrity.**

BUSINESS CASE

Measurable Business Impact & Stakeholder ROI

Why institutional developers and financiers mandate Center Estates for project oversight.

For Real Estate Developers

• Complete visibility into actual daily site progress.

• 15%–20% reduction in unauthorized material overages.

• Seamless transition into Sales & Facilities modules upon project completion.

For Project Managers & QSs

• Automated Interim Payment Certificates (IPCs) in minutes instead of days.

• Standardized daily site logs and concrete test archives.

• Clear evidentiary defense in contractor dispute arbitration.

For Commercial Banks & Financiers

• Independent, unassailable audit proof for construction loan drawdowns.

• Cryptographic evidence eliminates contractor collusion risks.

• ESG and regulatory safety compliance verification.

AUDIT PRINCIPLE **Center Estates transforms construction monitoring from an administrative chore into a strategic financial safeguard.**

SUMMARY & CONCLUSION

One Unified Platform: From Groundbreaking to Estate Handover

Thank you. We are ready for live scenarios and questions.

RECORD → VERIFY ON-SITE → STAGE-GATE → CERTIFY PAYMENT → HANDOVER

The mobile app is live. The backend is synchronized. The construction module is ready.

Live Demonstration Access

Web: https://centerestates.com
Mobile: exp://vpofj68-anonymous-8081.exp.direct
Demo Project: Karam Apartments (PRJ-KRM-01)

Presenter Contact & Inquiries

Platform: Center Estates Construction Suite
Engineering: Antigravity AI Systems
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